

Douglas Alencar

DATA & AI ENGINEER

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[portfolio-douglas-alencar.web.app](#)

PROFESSIONAL SUMMARY

Data & AI Engineer with 8 years of experience in technology, combining data engineering, analytics, governance, and AI application development. Proven track record in demanding corporate environments across healthcare, industry, and applied research, focused on turning data and models into usable, traceable, production-ready solutions. Hands-on experience with Python, SQL, PostgreSQL, GCP, BigQuery, FastAPI, and generative AI applications using RAG, agents, LangChain, LangGraph, and LLM integration.

CORE COMPETENCIES

AI Applications

RAG, embeddings, semantic search, LangChain/LangGraph, agents, and evaluation.

Data Engineering

Python, SQL, PostgreSQL, BigQuery, pipelines, and cloud.

Backend & APIs

FastAPI, Pydantic, SQLAlchemy, REST, and Docker.

Cloud & Analytics

GCP, Azure, Fabric, Power BI, and enterprise platforms.

AI-assisted Engineering

SDD, harness, Codex, Claude Code, Git, testing, and CI/CD.

Governance & Production

Architecture, security, observability, quality, and deployment.

PROFESSIONAL EXPERIENCE

- A.C.Camargo Cancer Center** | Data, Analytics, Governance & AI

3 years

 - Managed and coordinated data, analytics, and AI initiatives in a highly complex oncology healthcare environment.
 - Led data governance, connecting business needs, data quality, architecture, and responsible data use.
 - Worked with the GCP data ecosystem, including BigQuery, pipelines, processing services, and data governance capabilities.
- AB InBev** | Technology & Global Corporate Environment

2.5 years

 - Worked in IT within a large-scale corporate environment, strengthening experience with data, enterprise systems, processes, and cross-functional collaboration.
 - Worked in an international, execution-driven environment, communicating with diverse technical and business stakeholders.
- Shell + UFRJ** | Engineering, Technology & Applied Research

2.5 years

 - Worked in a university-industry partnership combining engineering, technology, and applied research in a multidisciplinary environment.
 - Developed experience in structured problem solving, technical documentation, and collaboration with multiple stakeholders.

EDUCATION & INTERNATIONAL EXPERIENCE

Postgraduate Studies in Engineering - UFRJ / UFRRJ • Science Without Borders - 18-month international exchange • Advanced English

TECHNICAL STACK

Data: Python • SQL • PostgreSQL • BigQuery • Spark/PySpark • dbt • Pandas • Polars
AI: RAG • Embeddings • Semantic Search • LangChain • LangGraph • MCP • LangSmith • AI Agents • Structured Outputs
Backend: FastAPI • Pydantic • SQLAlchemy • REST • Alembic • Docker
Cloud & Analytics: GCP • Azure • Fabric • Power BI • SAP
Engineering: Git • SDD • Harness • Codex • Claude Code • Testing • CI/CD • Observability

SELECTED PROJECTS

AI Customer Service & Scheduling Platform | Public case study in production

In production

Demonstrate AI customer service and scheduling application deployed to production with a fully simulated domain and dataset.

- Combines RAG, LLMs, relational and vector data, customer service workflows, and multiple levels of autonomy.
- Modular monolith: Python/FastAPI/PostgreSQL backend and React/TypeScript frontend.
- Production on GCP with Cloud Run, Cloud Build, Artifact Registry, Secret Manager, and Firebase; PostgreSQL on Neon.
- V1 delivered in 15 days with SDD, Codex, and Claude Code: 69 commits, 473 tasks, 28 migrations, 247 tests, and 17 Playwright scenarios.

Live application: plataforma-atendimento-prod.web.app

Data Change Risk Analyst | AI for data change governance

In production

Corporate AI tool that assesses whether changes to data assets can be made safely by combining evidence collection, a deterministic risk policy, a non-binding LLM recommendation, and human review.

- LangGraph for orchestration, fan-out/fan-in, checkpointing, and interrupt/resume; bounded ReAct agent for evidence gaps.
- LangChain for structured outputs and tools; MCP for optional downstream-usage reads; LangSmith for tracing.
- Real PostgreSQL introspection and deterministic Python risk rules; the LLM never decides the category or routing.
- Production on Cloud Run + Neon + Firebase/Secret Manager. V1 in 5 days: 31 commits, 76 tests, and 21 ADRs.

Live application: analizador-de-risco.web.app

Data Engineering on GCP | Portfolio project

In development

- Project designed to demonstrate pipelines, processing, modeling, data quality, and data delivery on Google Cloud.
- Complements the AI case study and reinforces the hybrid Data & AI Engineer positioning.

Data & AI Engineer Portfolio | Personal product

In production

- Bilingual web resume in production with static architecture, light/dark themes, case studies, and low-friction contact options.
- Next.js, React, TypeScript, Tailwind CSS, Firebase Hosting, and CI/CD on GitHub.

PROFESSIONAL DIFFERENTIATORS

- Combines corporate data/governance experience with end-to-end AI application development.
- Able to engage with business stakeholders, design architecture, and drive implementation through testing and production.
- Uses coding agents within a controlled engineering process with specifications, traceability, and validation - not just ad hoc code generation.

LEARN MORE

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